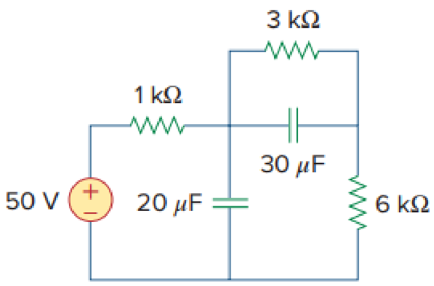
Chapter 6, Problem 5PP

Problem

Under dc conditions, find the energy stored in the capacitors in Fig. 6.13.

Fig. 6.13:



Step-by-step solution

Step 1 of 4

Consider the following *RC* circuit:

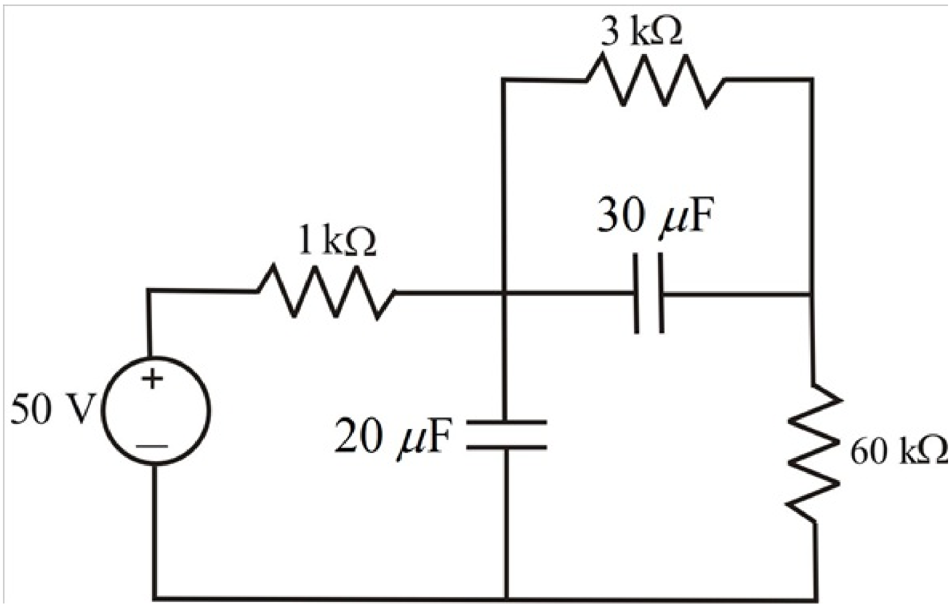


Figure 1

Step 2 of 4

Under the dc conditions, capacitor acts as an open circuit. Redraw the circuit in Figure 1 by replacing the two capacitors with an open circuit.

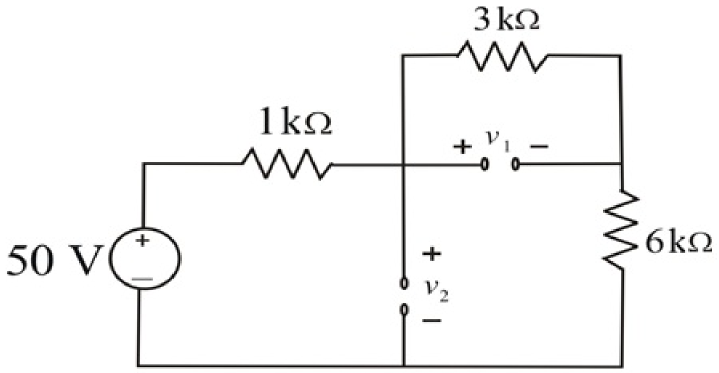


Figure 2

Apply Kirchhoff's voltage law to the closed loop of the circuit in Figure 2.

$$-50 + i\times 1\times 10^3 + i\times 3\times 10^3 + i\times 6\times 10^3 = 0$$
$$i\times 10\times 10^3 = 50$$
$$i = \frac{50}{10\times 10^3}$$
$$i = 5\text{ mA}$$

Determine the voltage v_1 across $3\text{ k}\Omega$ resistor by using Ohm's law.

$$\begin{aligned} v_1 &= Ri \\ &= (3\times 10^3)(5\times 10^{-3}) \\ &= 15\text{ V} \end{aligned}$$

Determine the voltage v_2 by applying Kirchhoff's voltage law to the loop.

$$-50 + (1\times 10^3)(5\times 10^{-3}) + v_2 = 0$$
$$-50 + 5 + v_2 = 0$$
$$v_2 = 45\text{ V}$$

Step 3 of 4

Determine the energy stored in the capacitor C_1.

$$\begin{aligned} w_1 &= \frac{1}{2}C_1 v_1^2 \\ &= \left(\frac{1}{2}\right)(30\times 10^{-6})(15^2) \\ &= 3.375\times 10^{-3}\text{ J} \\ &= 3.375\text{ mJ} \end{aligned}$$

Thus, the energy stored in the capacitor C_1 is $\boxed{3.375\text{ mJ}}$.

Step 4 of 4

Determine the energy stored in the capacitor C_2

$$\begin{aligned} w_2 &= \frac{1}{2}C_2 v_2^2 \\ &= \left(\frac{1}{2}\right)(20\times 10^{-6})(45^2) \\ &= 20.25\times 10^{-3}\text{ J} \\ &= 20.25\text{ mJ} \end{aligned}$$

Thus, the energy stored in the above capacitor C_2 is $\boxed{20.25\text{ mJ}}$.